Cambridge (CIE) A Level Maths: Pure 1

Your notes

Transformations of Functions

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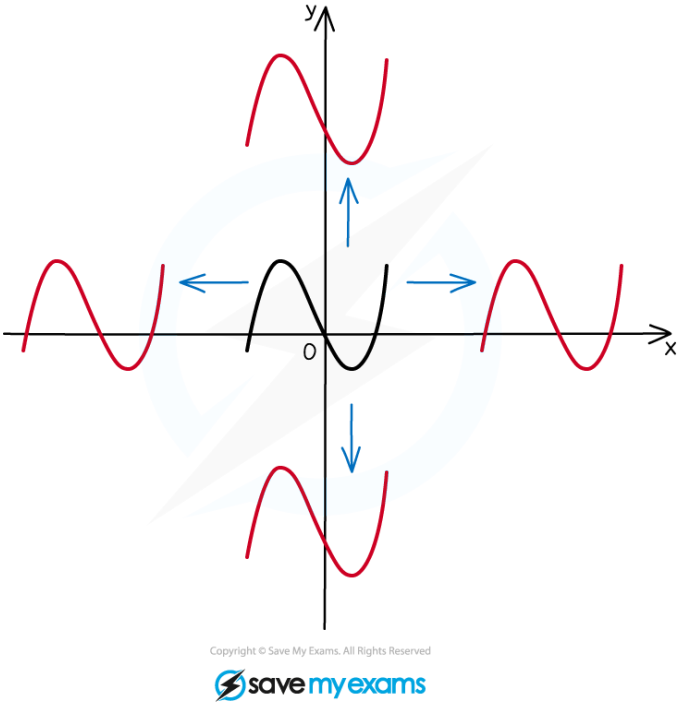
Your notes

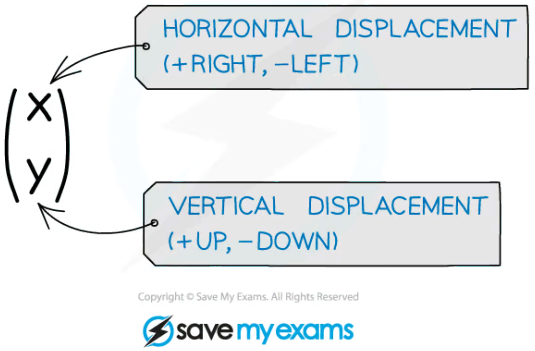
Translations

What are graph translations?

- When you alter a function in certain ways, the effects on the graph of the function can be described by geometrical transformations
- With a **translation** the shape, size, and orientation of the graph remain unchanged – the graph is merely shifted (up or down, left or right) in the *xy* plane



- A particular translation (how far left/right, how far up/down) is specified by a **translation vector**:



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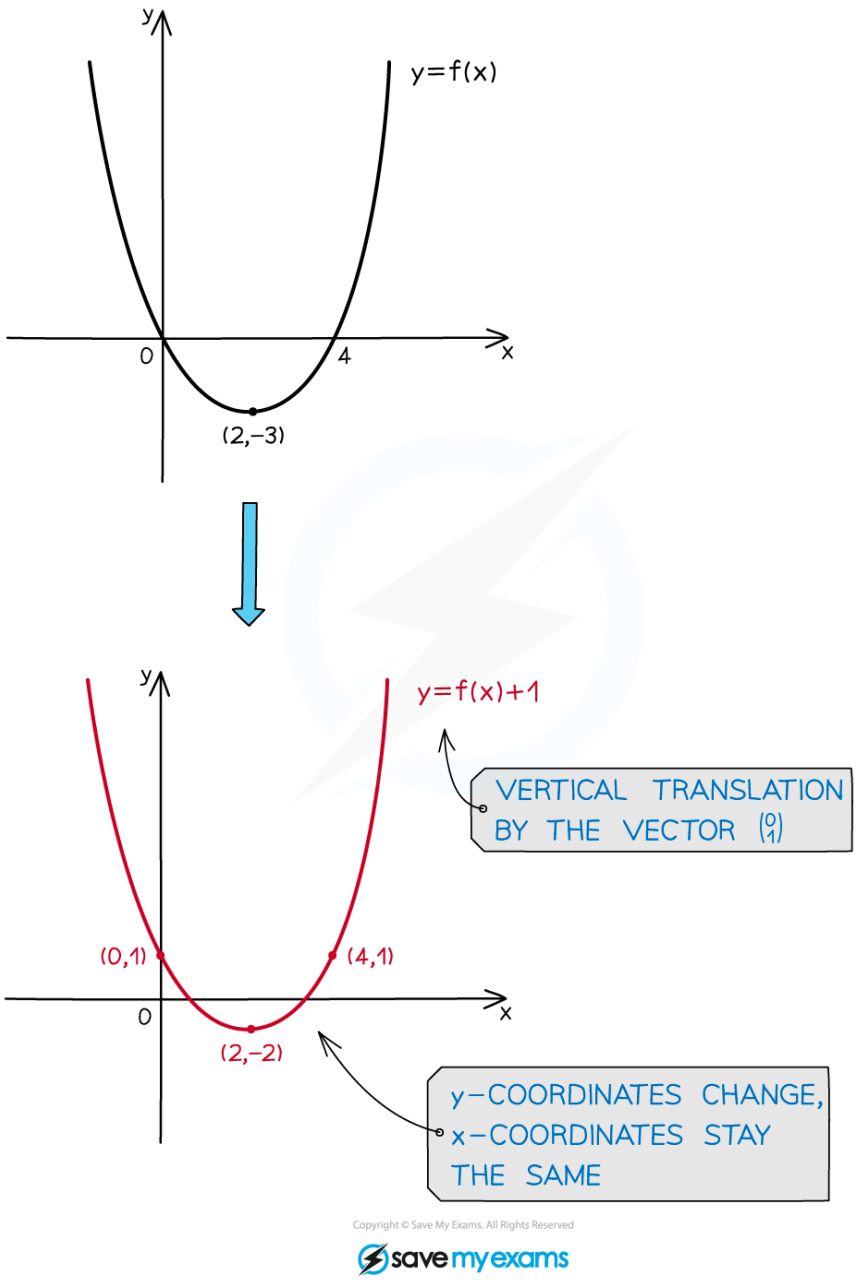
What do I need to know about graph translations?

- The graph of $y = \mathrm{f}(x) + a$ is a **vertical** translation of the graph $y = \mathrm{f}(x)$ by the vector $\begin{pmatrix} 0 \\ a \end{pmatrix}$
  - The graph moves **up for positive** values of *a* and **down for negative** values of *a*
  - The *x*-coordinates stay the same



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- The graph of $y = \mathrm{f}(x + a)$ is a **horizontal** translation of the graph $y = \mathrm{f}(x)$ by the vector $\begin{pmatrix} -a \\ 0 \end{pmatrix}$
  - The graph moves **left for positive** values of *a* and **right for negative** values of *a*
  - The *y*-coordinates stay the same

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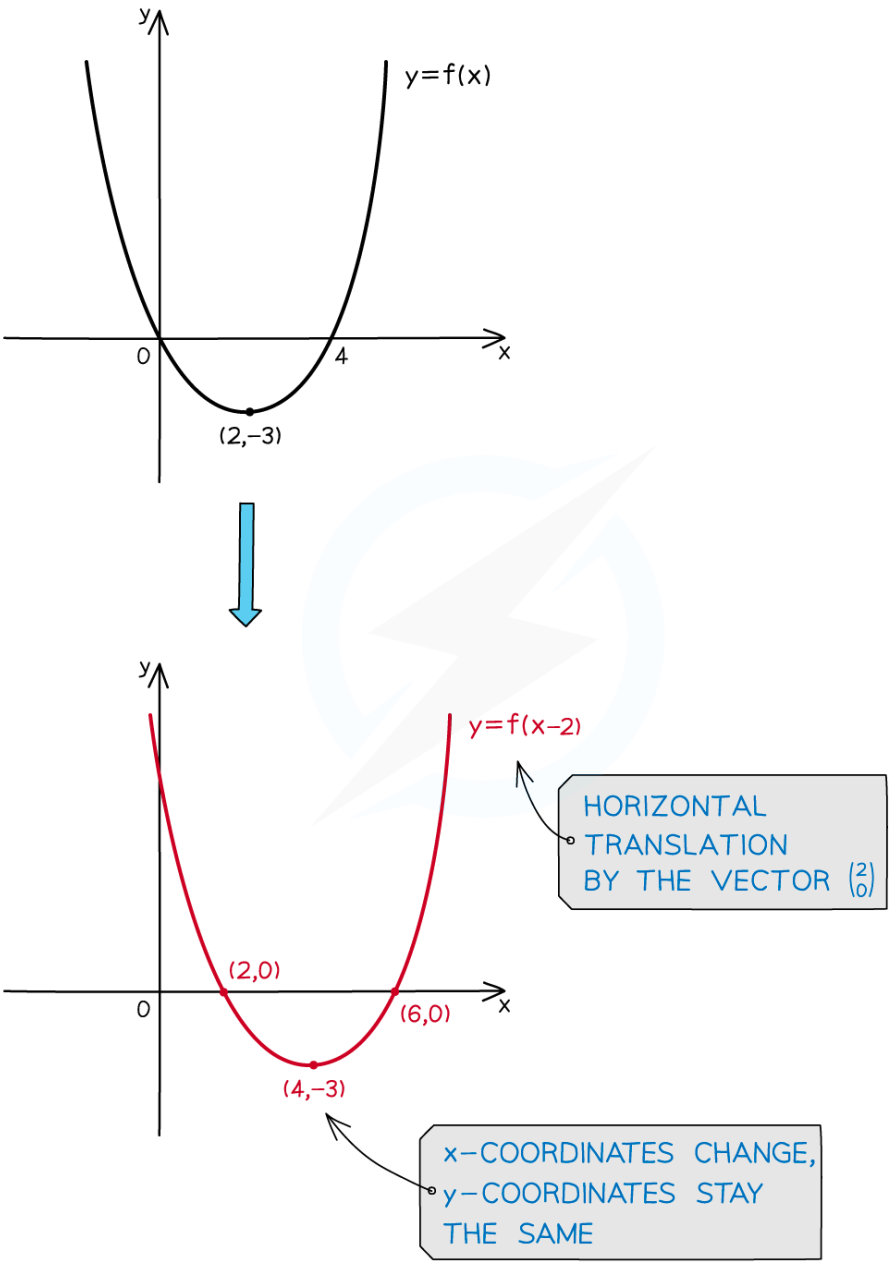

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- Any asymptotes of **f(x)** are also translated. If an asymptote is parallel to the direction of translation, however, it will not be affected

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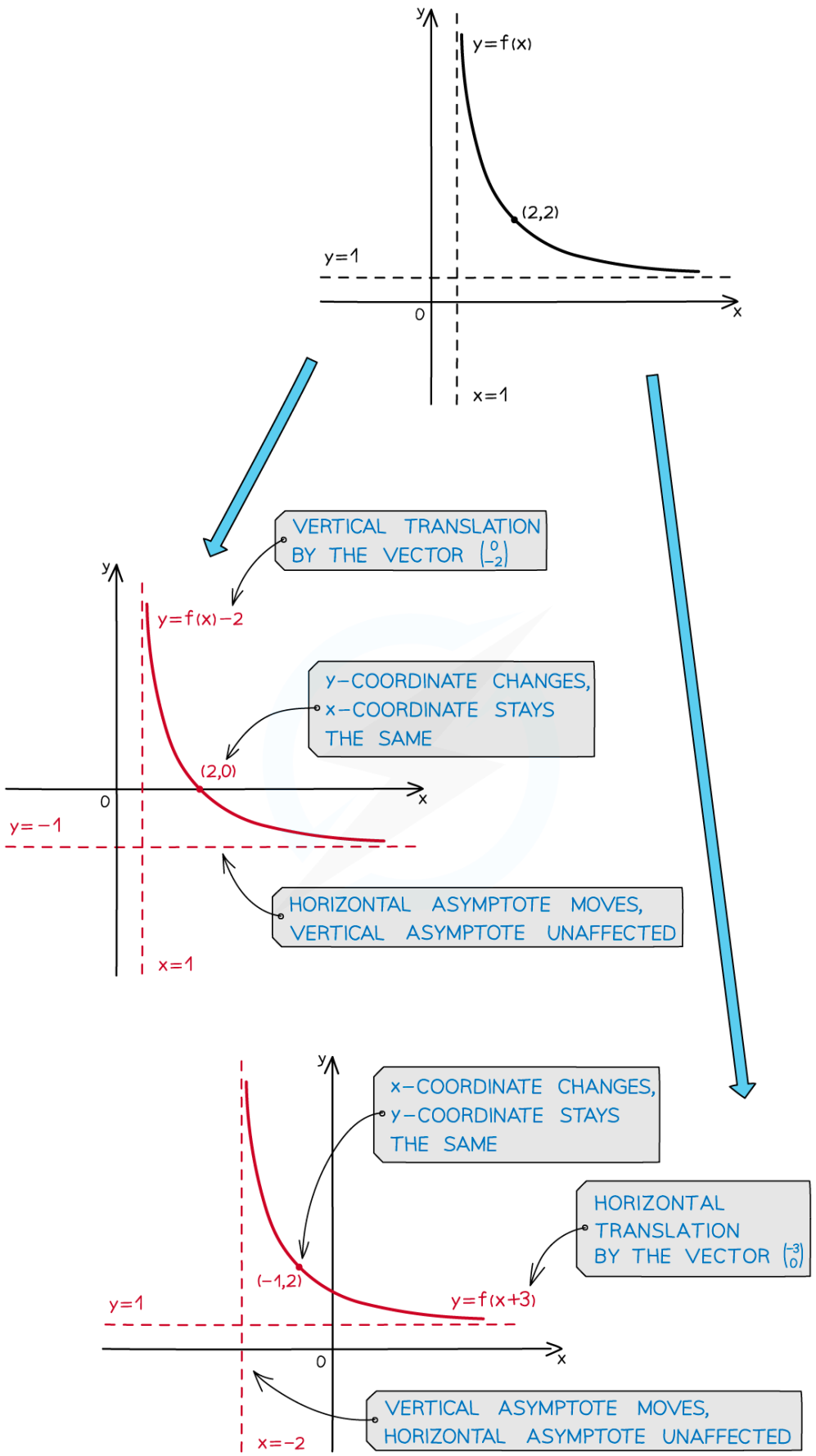
y
y=f(x)
(2,2)
y=1
0
x
x=1
VERTICAL TRANSLATION BY THE VECTOR $\begin{pmatrix}0\\-2\end{pmatrix}$
y
y=f(x)−2
y−COORDINATE CHANGES, x−COORDINATE STAYS THE SAME
(2,0)
0
x
y=−1
HORIZONTAL ASYMPTOTE MOVES, VERTICAL ASYMPTOTE UNAFFECTED
x=1
y
x−COORDINATE CHANGES, y−COORDINATE STAYS THE SAME
HORIZONTAL TRANSLATION BY THE VECTOR $\begin{pmatrix}-3\\0\end{pmatrix}$
(−1,2)
y=1
y=f(x+3)
0
x
VERTICAL ASYMPTOTE MOVES, HORIZONTAL ASYMPTOTE UNAFFECTED
x=−2

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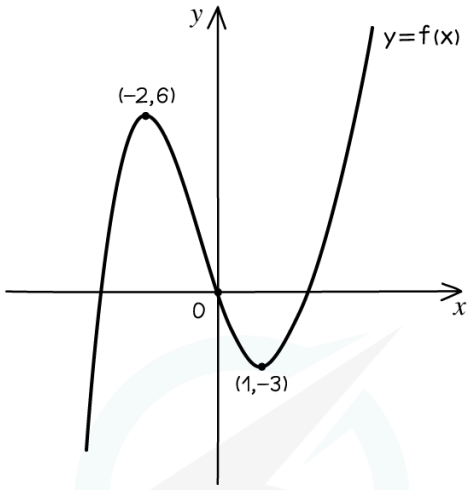
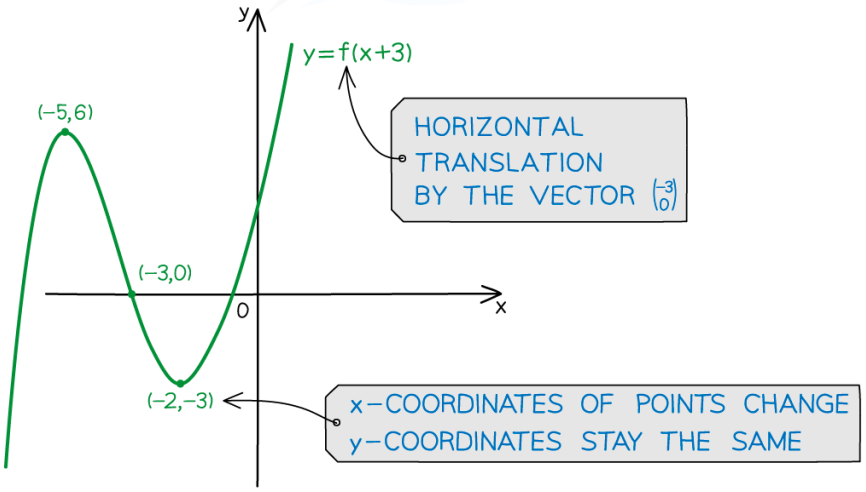
Your notes

Worked Example

The diagram below shows the graph of $y = f(x)$.



Sketch the graph of $y = f(x + 3)$.



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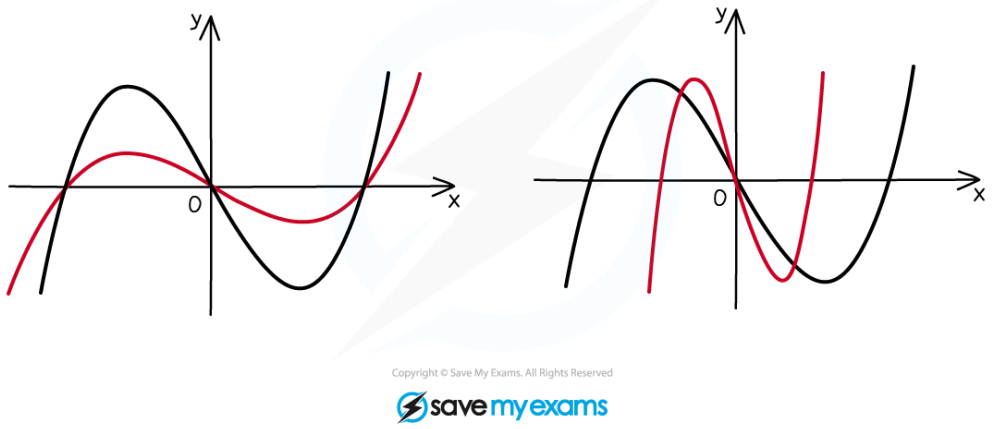
Stretches

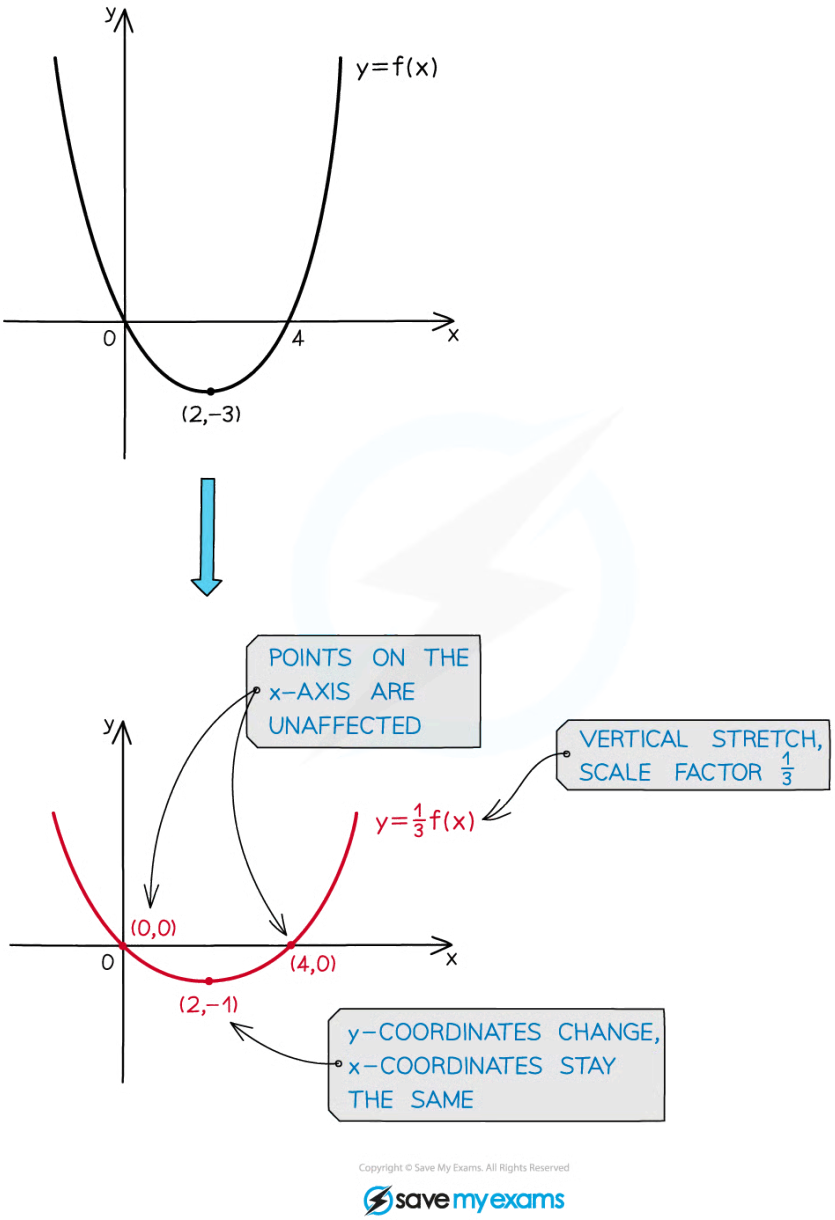
What are graph stretches?

- When you alter a function in certain ways, the effects on the graph of the function can be described by geometrical transformations
- With a **stretch** all the points on the graph are moved towards or away from either the ***x*** or the ***y*** axis by a constant scale factor

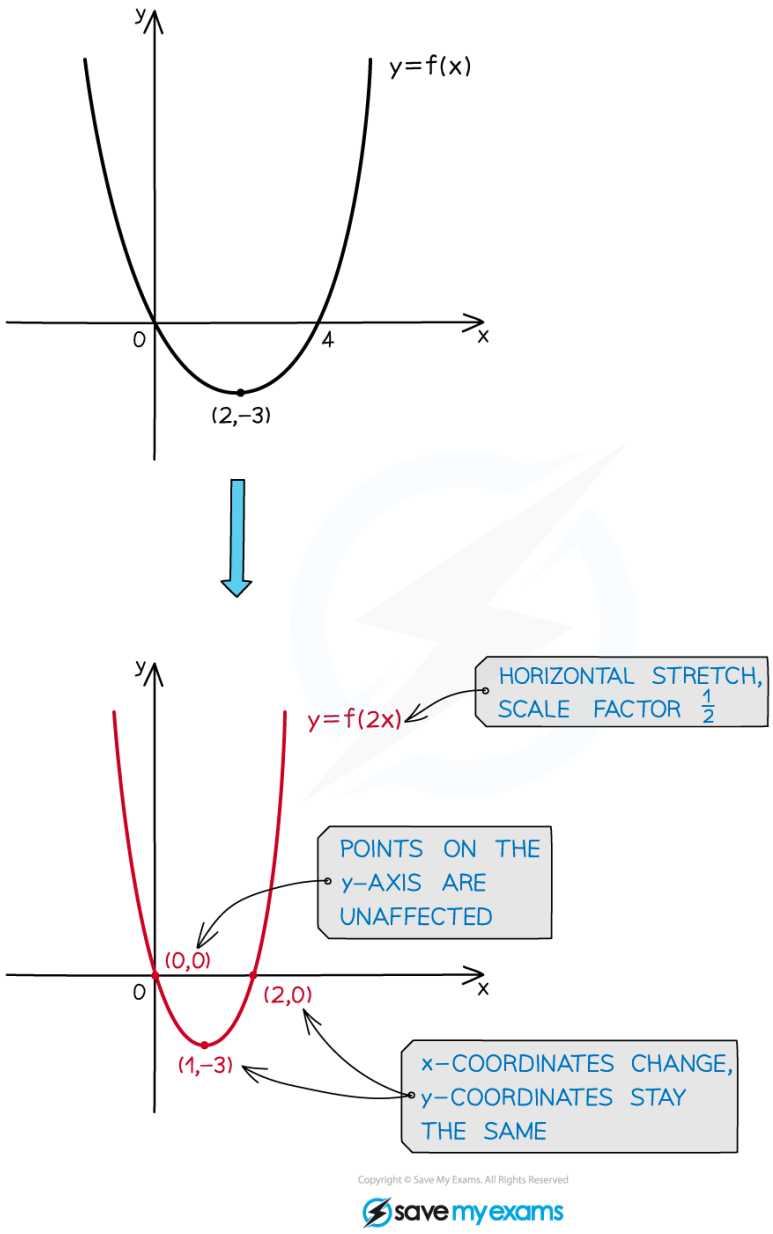


What do I need to know about graph stretches?

- The graph of ***y* = *a*f(*x*)** is a **vertical** stretch of the graph ***y* = f(*x*)** by a scale factor of ***a***, centred on the ***x*** axis
  - The ***x*** coordinates of points stay the same; ***y*** coordinates are multiplied by ***a***
  - Points **on the *x* axis** stay where they are
  - All other points move **parallel to the *y* axis**, away from (***a* > 1**) or towards (**0 < *a* < 1**) the *x* axis

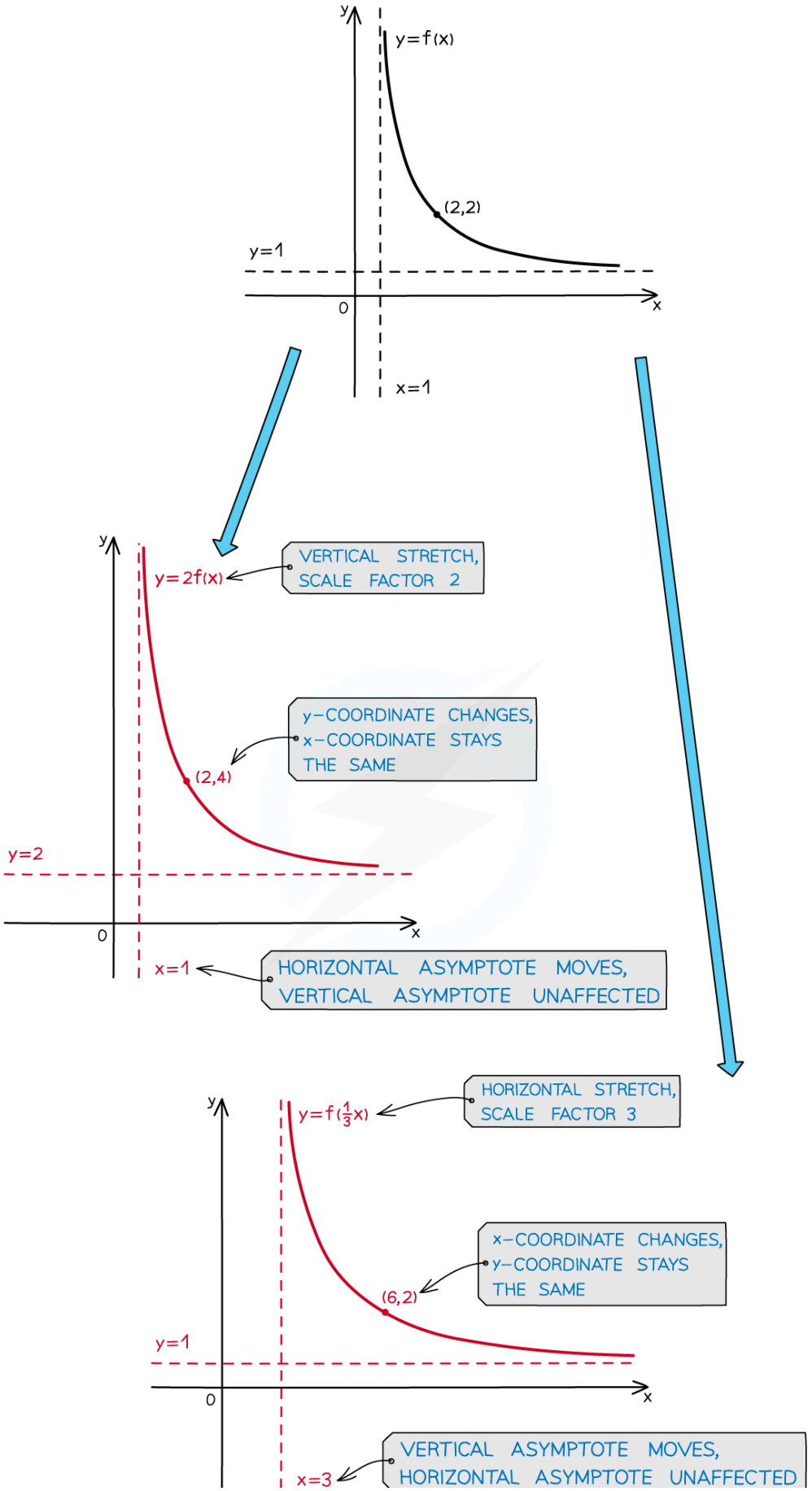


- The graph of ***y* = f(*ax*)** is a **horizontal** stretch of the graph ***y* = f(*x*)** by a scale factor of $\frac{1}{a}$, centred on the *y*
  - The ***y*** coordinates of points stay the same; ***x*** coordinates are multiplied by $\frac{1}{a}$
  - Points **on the *y* axis** stay where they are
  - All other points move **parallel to the *x* axis**, away from (**0 < *a* < 1**) or towards (***a* > 1**) the *y* axis



- Any asymptotes of **f(x)** are also affected by the stretch (stretch them as you would stretch the function of a straight line)
- If an asymptote is one of the coordinate axes, or is parallel to the direction of the stretch, however, it will not be affected

Your notes


y
y=f(x)
(2,2)
y=1
0
x
x=1
y
y=2f(x)
VERTICAL STRETCH,
SCALE FACTOR 2
y-COORDINATE CHANGES,
x-COORDINATE STAYS
THE SAME
(2,4)
y=2
0
x
x=1
HORIZONTAL ASYMPTOTE MOVES,
VERTICAL ASYMPTOTE UNAFFECTED
y
y=f(1/3 x)
HORIZONTAL STRETCH,
SCALE FACTOR 3
x-COORDINATE CHANGES,
y-COORDINATE STAYS
THE SAME
(6,2)
y=1
0
x
x=3
VERTICAL ASYMPTOTE MOVES,
HORIZONTAL ASYMPTOTE UNAFFECTED

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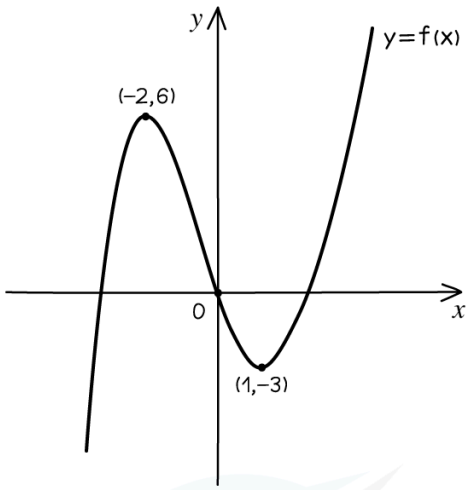
Examiner Tips and Tricks

- When you sketch a stretched graph, be sure to indicate the new coordinates of any points that are marked on the original graph.
- Try to indicate the coordinates of points where the stretched graph intersects the coordinate axes (if you don't have the equation of the original function this may not be possible).
- If the graph has asymptotes, don't forget to sketch the asymptotes of the stretched graph as well.

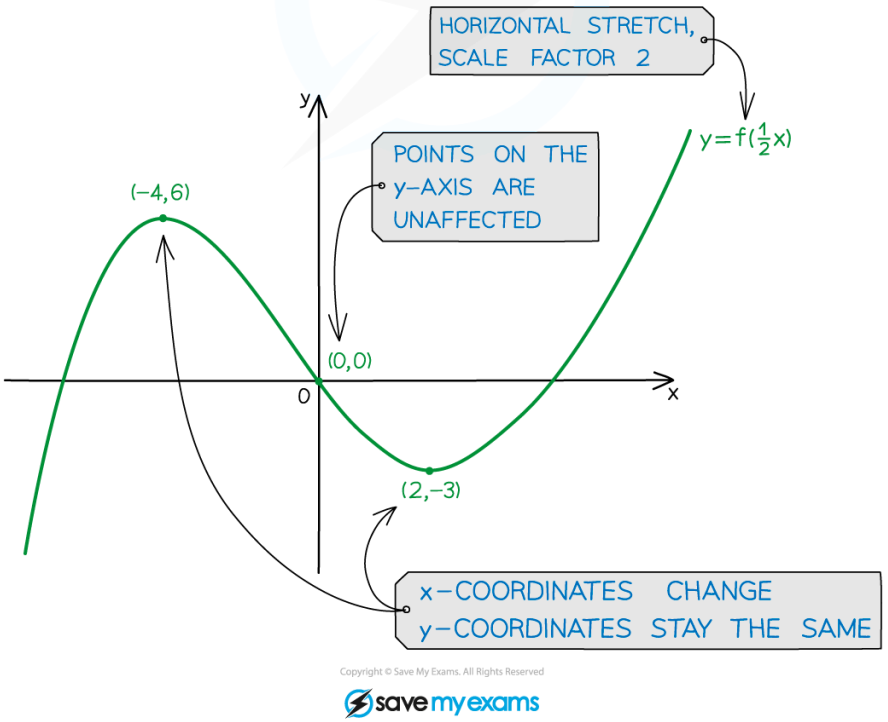
Worked Example

Your notes

The diagram below shows the graph of $y = f(x)$.



Sketch the graph of $y = f(\frac{1}{2}x)$.



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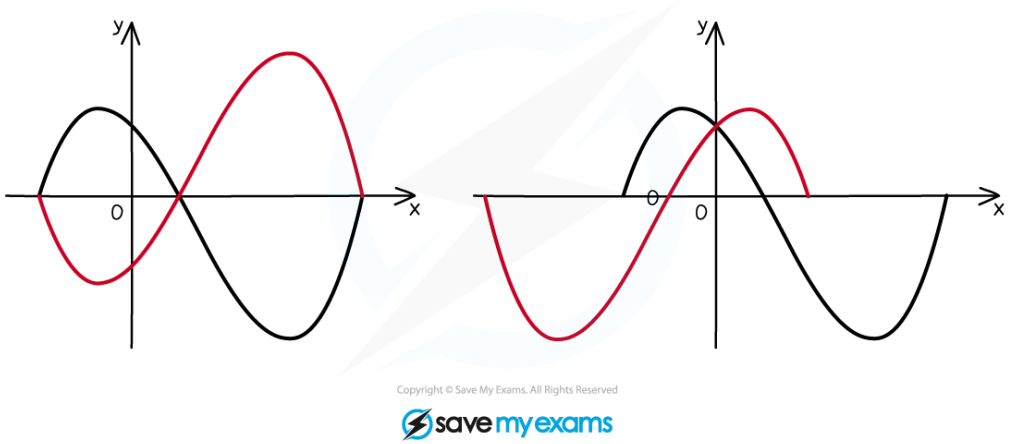
Your notes

Reflections

What are graph reflections?

- When you alter a function in certain ways, the effects on the graph of the function can be described by geometrical transformations
- With a **reflection** all the points on the graph are reflected in either the ***x*** or the ***y***

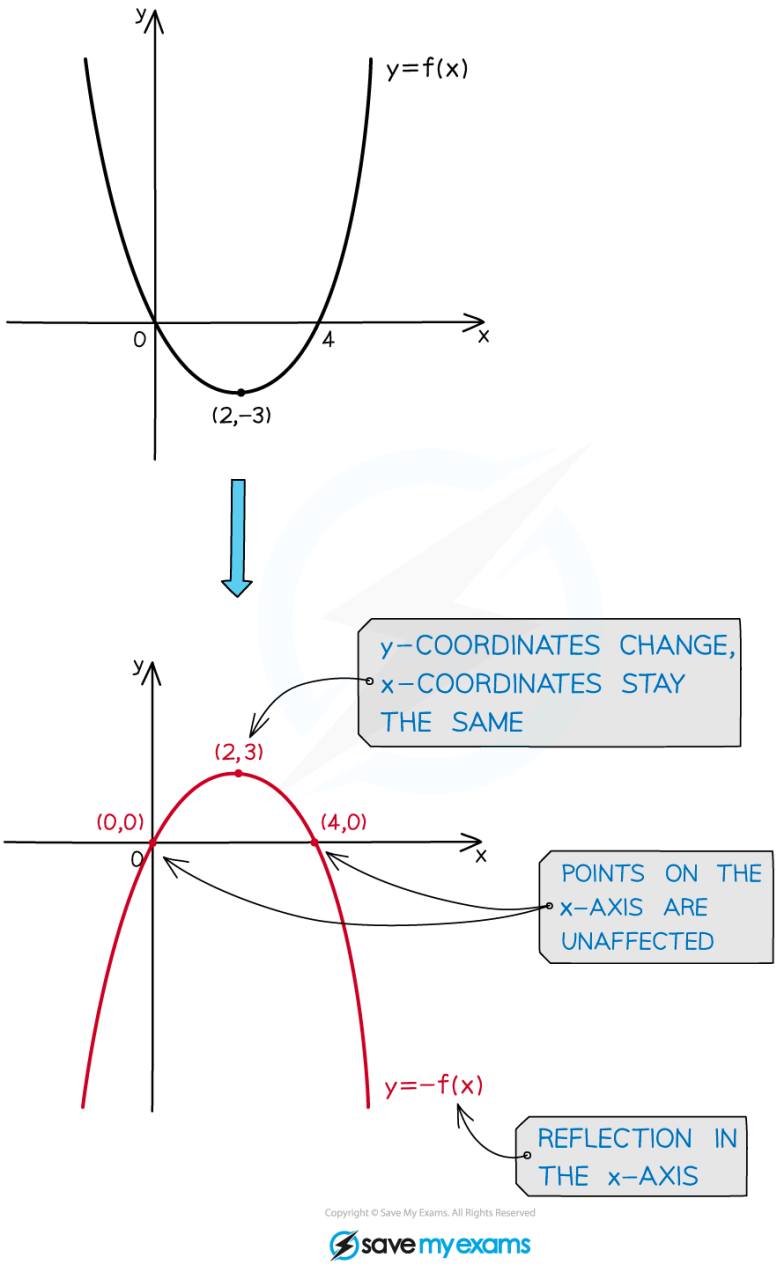


- Any asymptotes of **f(*x*)** are also affected by the reflection (reflect them as you would reflect the function of a straight line)
- If an asymptote is one of the coordinate axes, or is perpendicular to the coordinate axis in which the graph is reflected, it will not be affected

What do I need to know about graph reflections?

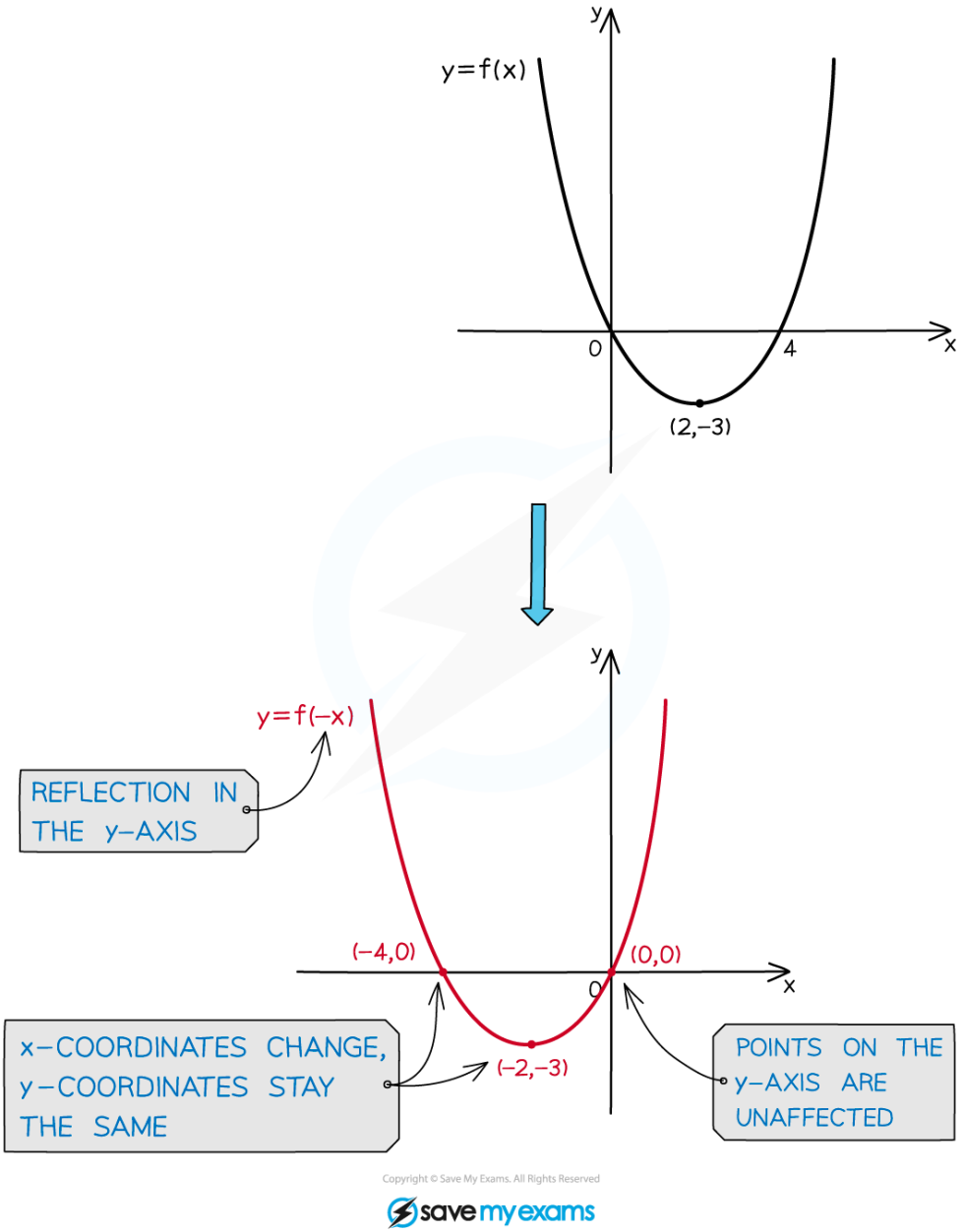
- The graph of ***y* = −f(*x*)** is a reflection in the *x* axis
  - The ***x*** coordinates of points stay the same; ***y*** coordinates have their signs flipped (positive to negative, negative to positive)
  - Points **on the *x* axis** stay where they are
  - All other points are reflected to the other side of the ***x*** axis

Your notes



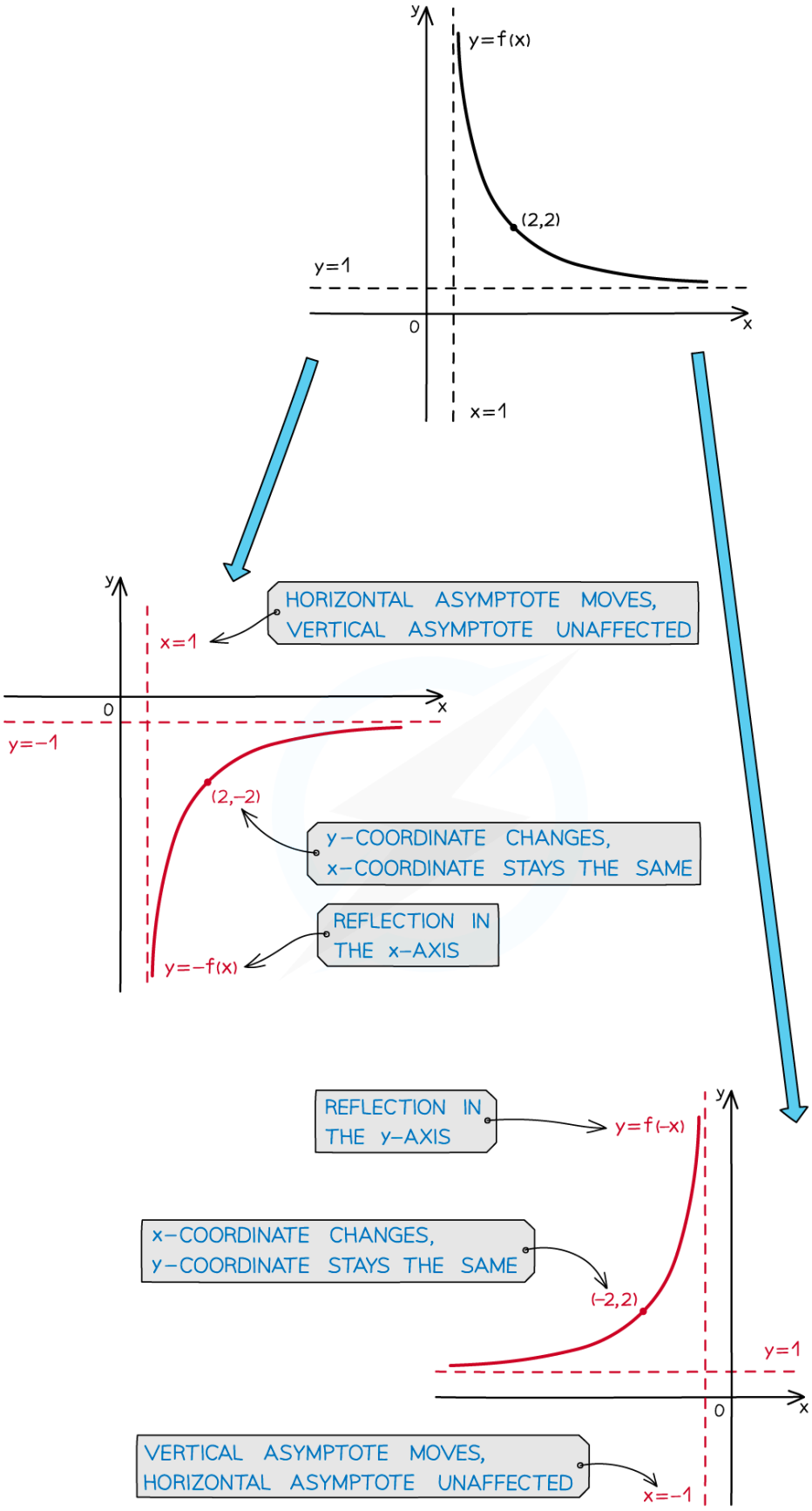
- The graph of $y = f(-x)$ is a reflection in the y axis
  - The y coordinates of points stay the same; x coordinates have their signs flipped (positive to negative, negative to positive)
  - Points **on the y axis** stay where they are
  - All other points are reflected to the other side of the y axis

Your notes



- Any asymptotes of **f(*x*)** are also affected by the reflection (reflect them as you would reflect the function of a straight line)
- If an asymptote is one of the coordinate axes, or is perpendicular to the coordinate axis in which the graph is reflected, it will not be affected

Your notes



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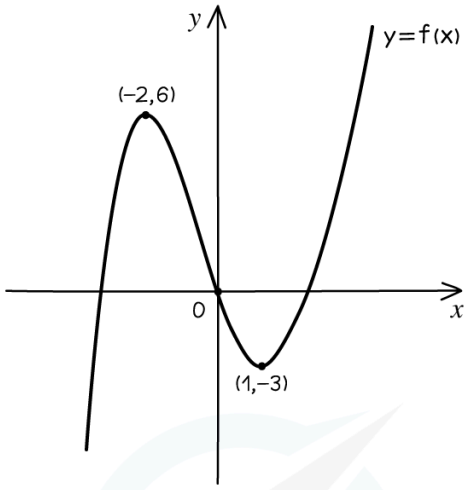
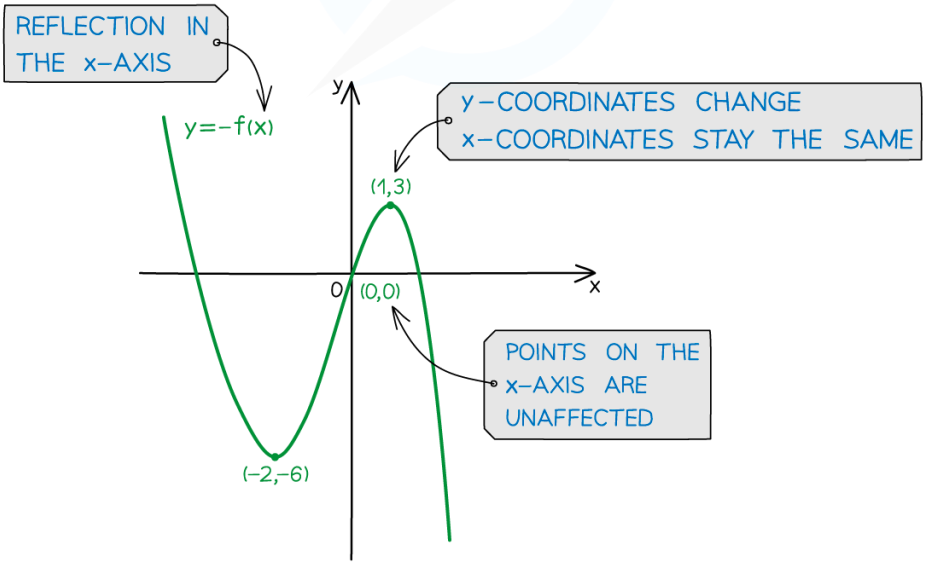
Worked Example

The diagram below shows the graph of $y = f(x)$.



Sketch the graph of $y = -f(x)$.



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